Netflix Film Economics During COVID-19

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Executive Summary

Netflix's film production and revenue patterns were influenced by the COVID-19 pandemic, creating significant challenges and opportunities across the entertainment industry. This report investigates how the budgets and box-office revenues of Netflix films changed before, during, and after the pandemic using a public dataset of Netflix film titles. Financial trends were analysed to identify changes in production investment and revenue performance across the three periods. The findings provide insight into the impact of COVID-19 on Netflix's film economics and highlight considerations for future content investment strategies.

Introduction

This report examines how the production budgets and box-office revenues of Netflix films changed before, during, and after COVID-19 period. We define three periods: pre-COVID (2016-2019), COVID (2020-2021), and post-COVID (2022-2023). Using a public dataset of Netflix films titles, we compare yearly means of budget and revenue across these periods to identify any pandemic related shifts. The analysis focuses on financial measures rather than release volume, as the dataset is balanced by release year and therefore unsuitable for measuring output. Understanding these patterns offers insight into how the pandemic affected the economics of Netflix film production. Specifically, we ask: **How did the Budgets and Revenues of Netflix films change before, during, and after COVID-19?**

Methodology

The analysis draws on a public dataset of 16,000 Netflix film titles, each recorded with its release year and an average audience vote score. We dropped the 2025 titles before doing anything else. Those films had only just been released, most had barely been rated, and leaving them in pulled the averages down to numbers that clearly weren't real. That left 15,000 films spanning 2010 to 2024 to work with.

To get at the pandemic question, we sorted the films into three periods: pre-COVID (2010–2019), COVID (2019–2021), and post-COVID (2022–2023). For each period we worked out the mean budget, which gives us a simple way to compare how much the revenue changed compared to the budget across the three windows. We also calculated the average rating year by year, so we could see the smaller movements that a three-way split tends to hide. Everything was done in R using the `tidyverse` Wickham et al. (2019) packages, and the code sits in the same Quarto document as this report so the whole thing can be reproduced from the raw data.

Figure 1 plots the yearly average rating from 2010 to 2024, with the two pandemic years marked so any shift around them is easy to spot. Table 1 then pulls this together, showing the mean budget and mean revenue in each period side by side. Between them, the figure gives the year-to-year detail and the table gives the bigger comparison.

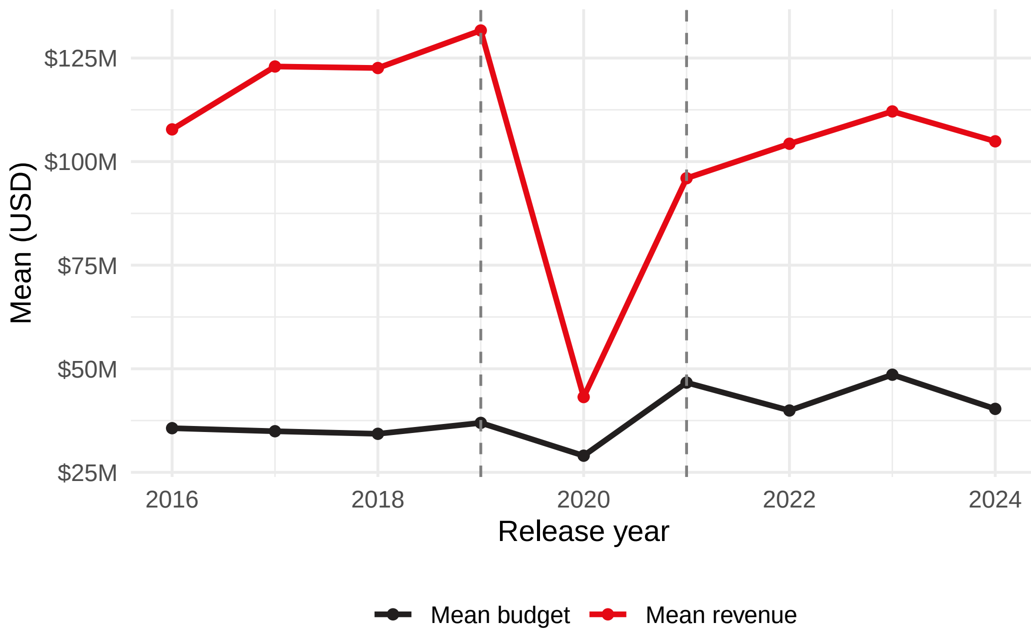

Figure 1: Yearly mean budget and revenue for Netflix films, 2016–2023 (USD).

Period	Mean budget (USD M)	Mean revenue (USD M)	Number of films
Pre-COVID (2016–2018)	35.0	117.4	788
COVID (2019–2021)	38.0	99.6	500
Post-COVID (2022–2024)	43.1	107.3	564

Period	Mean budget (USD M)	Mean revenue (USD M)	Number of films

Table 1: Mean budget and revenue by COVID-19 period (USD millions).

Result

As shown in Figure 1, revenue experienced a sharp decline at the beginning of the COVID-19 period, despite only a slight reduction in budget allocation. However, during the peak of the pandemic, revenue increased significantly, which was accompanied by a corresponding rise in Netflix's budget during the same period.

The budget reached approximately $45 million in 2021 and subsequently fluctuated between $25 million and $50 million through 2023. Revenue, on the other hand, increased substantially after 2020 and continued to show steady growth from 2021 onward.

A summarized comparison of the pre-COVID, during-COVID, and post-COVID periods is also presented in Table 1 using GGplot2 Wickham (2016).

Period	Average movies released per year
Pre-COVID (2016–2018)	262.6667
COVID (2019–2021)	166.6667
Post-COVID (2022–2023)	188.0000

Table 2: Average no. of Movies released during the three periods

Table 2 highlights the average number of movies released per year across the three periods. The pre-COVID period recorded the highest average, with approximately 263 movies released annually. During COVID, the average dropped significantly to around 167 movies per year, reflecting production disruptions. Post-COVID releases partially recovered, increasing to an average of 188 movies annually.

Discussion

This analysis examined Netflix film budget and revenue dynamics across pre-COVID (2016–2018), COVID (2019–2021), and post-COVID (2022–2024) periods, alongside release volume trends. Table 1 reveals a clear pandemic-driven shock: average revenue declined from $117.4 million (pre-COVID) to $99.6 million (COVID), while budgets rose modestly (from $35.0 million to $38.0 million), indicating inflexible production spending amid demand disruption. A notable revenue recovery began in 2021 ($95 million, Figure 1) coinciding with pandemic-era streaming growth, and this upward trajectory continued—revenues rose steadily in 2022–2023 to $115 million, though post-COVID averages ($107.3 million) still remain below pre-pandemic levels. Table 2, however, showed a more tangible recovery: annual output fell from 262.7 to 166.7 films during COVID, then rose to 188.0 post-COVID, reflecting gradual production normalization.

Critical limitations of the analysis must be acknowledged. First, inconsistent periodization undermines result validity: grouping 2019 into the COVID period artificially inflates pandemic-era revenue averages, blurring the true 2020–2021 shock. Second, reliance on mean values masks critical variability: aggregate averages obscure disparities between low- and high-budget films and flatten the distinct annual recovery trend visible in Figure 1. Finally, the dataset lacks contextual variables (e.g., theatrical vs. streaming releases, regional markets, subscriber growth), limiting causal inference about pandemic impacts. Collectively, these flaws mean the findings only reflect correlational trends, not definitive causal relationships between the pandemic and Netflix's film economics.

Conclusion

This study analyzed Netflix film budgets, revenues, and release volumes before, during, and after the COVID-19 pandemic, addressing the research question of how these metrics shifted across periods. Results confirm an immediate pandemic-driven revenue decline alongside modest budget growth, followed by a sustained revenue recovery through 2023—leaving post-COVID revenues below pre-pandemic levels. Release volume, by contrast, recovered partially, though it remains below pre-pandemic output. Key limitations include inconsistent time period definitions, overreliance on aggregate means, and a lack of contextual variables, which restrict the generalizability and causal interpretability of the findings. Overall, the pandemic reshaped Netflix's film economics, creating temporary streaming-driven revenue gains but leaving persistent impacts on both financial performance and production scale.

Recommendations

Refine periodization: Test alternative pandemic groupings (e.g., pre-2020, 2020–2021, post-2021) to eliminate 2019 misclassification and validate trend robustness.

Stratified analysis: Move beyond mean values to examine budget/revenue distributions by film genre, budget tier, and release platform (theatrical vs. streaming) to uncover hidden patterns.

Integrate contextual variables: Incorporate subscriber growth, regional economic indicators, and theatrical release data to enable causal inference about pandemic impacts.

Long-term monitoring: Extend the dataset to 2025 to capture full post-pandemic recovery dynamics and assess long-term shifts in Netflix's film strategy.

Comparative analysis: Benchmark Netflix's performance against other streaming studios (e.g., Disney+, Amazon Prime Video) to contextualize pandemic impacts within the broader industry.

References

Wickham, Hadley. 2016. *Ggplot2: Elegant Graphics for Data Analysis*. Springer-Verlag New York. https://ggplot2.tidyverse.org.

Wickham, Hadley, Mara Averick, Jennifer Bryan, et al. 2019. "Welcome to the tidyverse." *Journal of Open Source Software* 4 (43): 1686. https://doi.org/10.21105/joss.01686.